

Block 1 Assignment: Submit One Mini-Design Challenge (Final)

Process Document

1. **Context and learners:** Who is it for, and what learning need does it address?

This project is design for middle school teachers and students taking part in a a PBL-style project called Our Golden Record. The project is based on the Golden Record project that was part of NASA's Voyager missions in the 1970s. The goal is for student groups to crate a modern version of this, utilizing modern media to represent their own communities.

The AI Council is available to students at various points of the project and simulates an advisory group or board. It gives teachers a tool for students to learn how consulting with such as group can improve their work, gaining valuable perspectives they not otherwise get. While teachers may arrange for in-person guest speakers, field trips, and connection to actual experts, the AI Council can fill in knowledge gaps and give students practice at using a AI-enabled tool.

2. **Design goals:** What were you trying to achieve (learning outcomes and/or experience goals)?

Learning goals for students:

Evaluate what defines your "community" by creating and selecting material that will be communicated.

Produce a range of media that communicates complex human concepts to an unfamiliar audience

Apply mathematical modeling to manage project specifications and constraints

Conduct sustained research into your community to present a multidisciplinary and multimodal digital artifact.

Demonstrate effective teamwork and project management skill in project groups to meet a launch deadline.

My personal goal for the project was to create a flexible for teachers that adds value to such a project, and is not AI/tech for the sake of AI/tech.

3. **Key design decisions:** What did you choose and why (e.g., scaffolds, feedback strategy, council roles and handoffs, modality choices)? What learning theories did you

apply and/or integrate into your design?

I scaffolded the release of the AI council members to align with different project phases. In practice, teachers would regulate student access to the AI council to particular times, such as one time during and at the conclusion of each project phases. This would include handoffs from the students to the council members, and can even include back and forth between council members.

I gave each council member prompting to be reflective; they do not merely provide answers, or even direct guidance, but ask open-ended follow-up questions that spur critical thinking and help students expand the scope of their research or presentations.

4. Learner interaction flow: How does a learner use this? What inputs do they provide and what do they get back?

Learners can provide either specific questions in a prompt box (eg. "How can we involve local indigenous history in our project?") or they can upload artifacts for the council to analyze and comment on, such as their Community Charter or their preliminary artifact list. They receive responses back from members, either one at a time, or at the same time. I have built-in accessibility features coded into my UI to address the age and grade-level of the students (aimed at 6-7th graders), including: larger font size for readability on screens, high-contrast for text and background, text output is throttled to approximately 175 wpm with pauses at the end of each sentence, spacing between paragraphs and questions, directions to aim for grade 6 lexile level for all output, including safeguards for language and content.

5. Iteration evidence: What feedback did you receive and what changed as a result?

I received feedback to consider interactions between council members. I have added a Send Response To... button after a user gets a response from a council member. This simulates interaction between the council. On the back-end I included an instruction for the receiving council member to analyze the response critically and provide either an alternate idea or one critique of the original response. Feedback has also motivated me to turn this into an AI Council template for teacher's to create custom personas and enter in their own project information.

6. Limitations and next steps: What's not working yet, what assumptions are you making, and what would you improve next iteration?

The interaction between council member has mixed results, and sometimes council members have stock answers that fall back to their original prompting. They also

sometimes refer to what should be teacher-facing information or prompts (specific standards or pedagogical language). I would address this by editing the council prompts further, and giving overall guidance to avoid such terms or topics.

I would also like to provide more student-facing information on the landing page, including project phase details, learning goals, or suggested prompts, without scaffolding too much. I also want to add reminders of the expertise of each council member, with overlays/bios or tooltips.

I also need to test with some project examples, either created by me or with another generative tool, such as NotebookLM.

7. Generative AI attribution: Tools used and a brief description of how they contributed?

To build my council, I created five separate personae on Google Gems that use Google Gemini. To create a simple UI, I used Cursor to build a web interface that puts each Gem on one screen with a prompt input. Because Gems cannot yet be embedded directly, or have multi-Gem capability on the same screen, the Gem's instructions are pasted into my web app's code, along with a Gemini API that uses Gemini Flash 2.5. As a result, the responses from the members are not as consistent and high quality as asking the Gems directly in the Google interface. To help with this, I added more detail into the context of the web app for each persona. This made up for some of the context that the Gems had within the Google interface (on Google platform, the Gems have access to a NotebookLM notebook I have that contains project documents and conversations about project planning). The web UI is hosted on a simple web app server I use for other projects, called Vercel and is backed up on a Github repository.